Changing velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate acceleration and distinguish speeding up, slowing down, and turning.

SUCCESS CRITERIA - I can...

1. Calculate acceleration from velocity change and time.
2. Compare acceleration direction with velocity direction.
3. Explain why turning at steady speed is acceleration.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “acceleration” and “initial” to explain your answer. Sentence starters are optional.

acceleration: rate of change of velocity

initial: at the beginning of the interval; subscript i

final: at the end of the interval; subscript f

01 Warm-up: read motion in two ways

(a) A remote-control car travels *west* at **4 m/s** for **6 s**. **How far *west* does it move?**
(b) A probe's velocity has components of **8 m/s** *east* and **3 m/s** *south* on a tracking map. **Sketch the components** and **the probe's overall direction**.

Sentence starter: In my work, I used … to show … . I labeled it … .

Marzano check - circle: ① ② ③

02 Worked example: braking on a bike

Reference: $a_{avg} = (v_f - v_i)/\Delta t$.
v_i and v_f are initial and final velocity. Δt is elapsed time. With a steady change in velocity, this gives the constant acceleration.

Worked reasoning: A cyclist brakes while riding *east*. Her speed drops steadily from **8 m/s** to **2 m/s** in **3 s**. Take *east* as positive.
The velocity change is **2** - **8** = **-6 m/s**.
Acceleration = **-6** ÷ **3** = **-2 m/s²**.
She is still moving *east*, but her acceleration points *west*.

Your turn: The **diagram** shows the cyclist’s velocity before and after braking.
1. **Draw an acceleration arrow** below the two velocity arrows.
2. Both velocity arrows point *east*. **Explain why the cyclist is slowing down.**

Sentence starter: I think … . I can point to …, which shows …, so … .



Acceleration: add a separate arrow.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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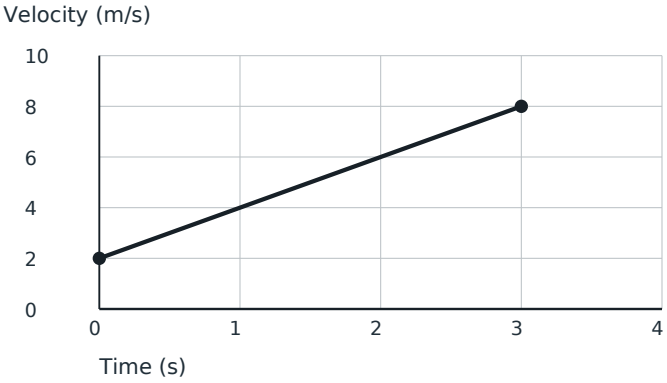
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 A remote-control car speeds up

The graph shows a remote-control car during a straight-track test. *East* is positive.
(a) **Read its velocity** at **0 s** and **3 s**.
(b) **Calculate the change in velocity**, then **the acceleration**.
(c) **What feature** of the graph shows that its speed is increasing?

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 Slowing before a crosswalk

A cyclist riding *east* slows steadily from **10 m/s** to **4 m/s** as she approaches a crosswalk. This takes **5 s**.
(a) **How much speed** does she lose each second?
(b) **Find her acceleration**, **including its direction**. Is she still moving *east*?

Support: Find the total speed lost first. Divide by the elapsed time. **Compare the direction of acceleration** with the direction of motion.

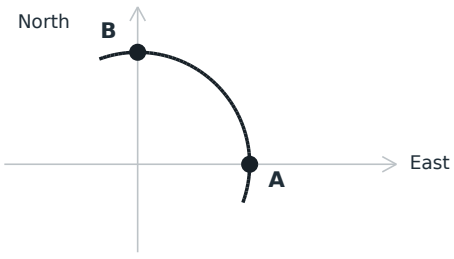
Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

05 Running around a bend

A runner goes from A toward B around a bend at a steady speed. A friend says, "If the speed stays the same, there is no acceleration."
Draw a velocity arrow at A and at B on the path below. **Do you agree** with the friend? **Use the arrows** to explain.

Sentence starter: I think … . I can point to …, which shows …, so … .



Path sketch: runner moves from A toward B.

Marzano check - circle: ① ② ③

Changing velocity

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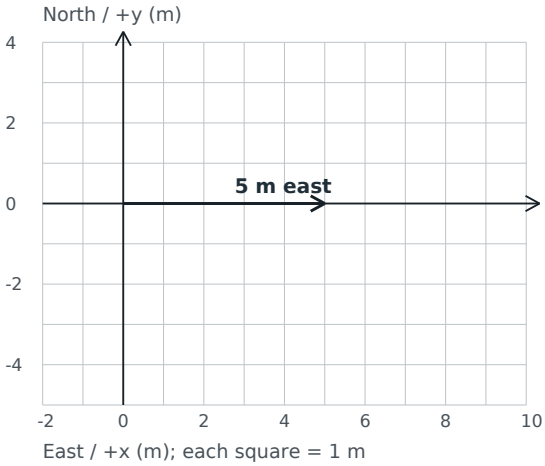
Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: a robot's route

During a navigation test, a robot drives **5 m** *east*, **3 m** *north*, **2 m** *west*, then **7 m** *south*.
(a) The first move is shown. **Complete the route** on the grid.
(b) **Draw an arrow** from the start to the endpoint. **How far *east*** or *west*, and *north* or *south*, does it finish from its start?

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

07 Exit check: a docking simulation

In a docking simulation, a spacecraft's velocity changes steadily over **2 s**. The arrows show its velocity at the start and end.
(a) **Read the speeds**. Is the spacecraft speeding up or slowing down?
(b) **Calculate its acceleration**. **Add an acceleration arrow** and **explain its direction**.

Sentence starter: I think … . I can point to …, which shows …, so … .



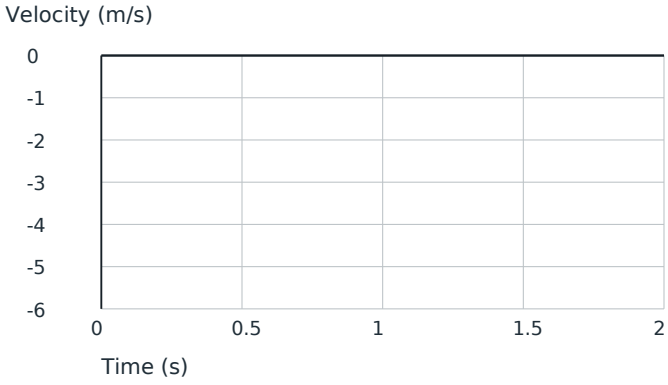
Acceleration: add a separate arrow.

Marzano check - circle: ① ② ③

08 STRETCH: negative can mean faster

In a straight-track test, a remote-control car speeds up steadily from **1 m/s** *west* to **5 m/s** *west* in **2 s**. *East* is positive.
Complete the velocity-time graph below. **Calculate acceleration** and **explain how a negative acceleration** can make this car speed up.

Sentence starter: I think … . I can point to …, which shows …, so … .



Marzano check - circle: ① ② ③